

CLUSTERING



K-means Algorithms &
the CURE Algorithm

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SUMMARY

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- The CURE Algorithm
- Conclusion



INTRODUCTION

Clustering is the process of examining a collection of “points” and grouping the points into “clusters” according to some distance measure.

It can be used :

- to identifies groups of consumers with the same behaviors
- for image analysis applications
- in bioinformatic to find genes, etc...



K-MEANS ALGORITHMS

K-means algorithms are the best known family of clustering algorithms. They assume a euclidean space and k-number of clusters.

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Initially choose k points that are likely to be in  
different clusters;  
Make these points the centroids of their clusters;  
FOR each remaining point p DO  
    find the centroid to which p is closest;  
    Add p to the cluster of that centroid;  
    Adjust the centroid of that cluster to account for p;  
END;
```



K-MEANS ALGORITHMS

To initialise the cluster we can :

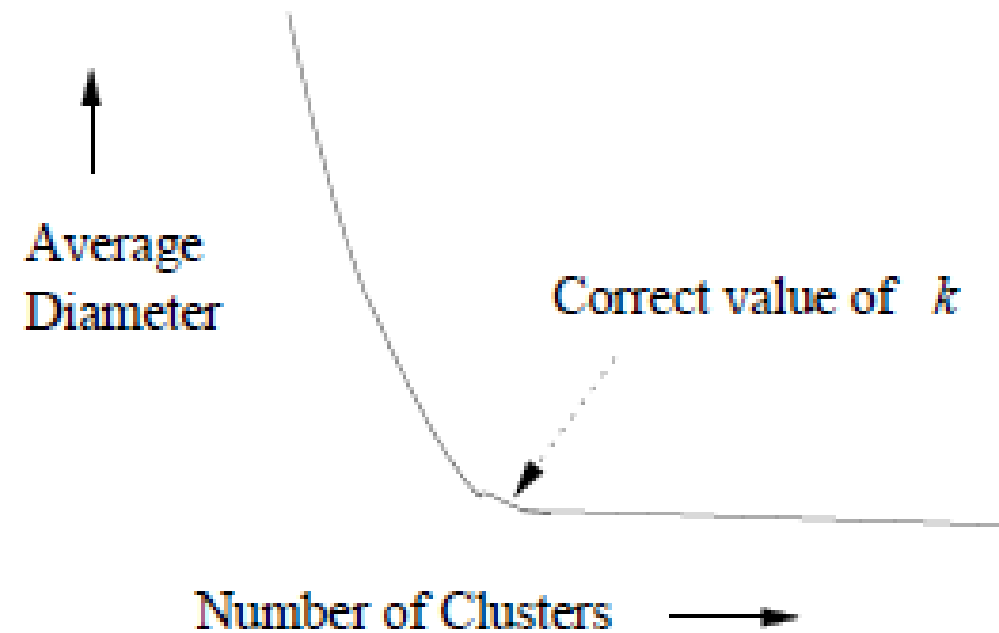
- Pick points that are as far away from one another as possible
- or
- Cluster a sample of the data so there are k -clusters and pick a point from each cluster



K-MEANS ALGORITHMS

Picking the right k-value :

if we take a measure of appropriateness for clusters, such as average radius or diameter, that value will grow slowly, as long as the number of clusters we assume remains at or above the true number of clusters. However, as soon as we try to form fewer clusters than there really are, the measure will rise precipitously.



Apparently, the right k-value can be found in a range $v/2$ and v that represents a very low average diameter modification.



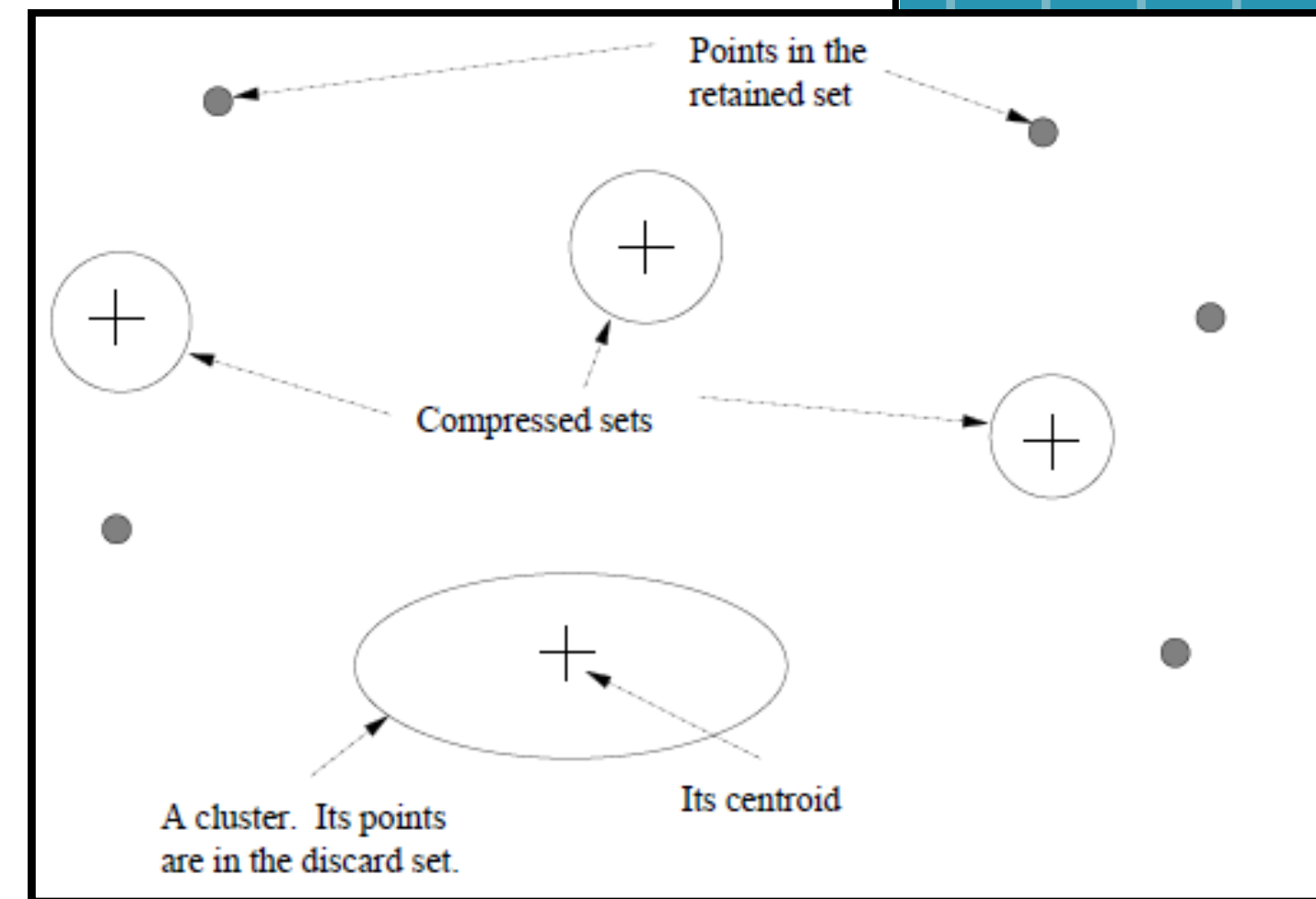
K-MEANS ALGORITHMS

The BFR Algorithm:

This is a variant of the k-means algorithm that is designed to cluster data in a high-dimensional Euclidean space. It makes an assumption that the clusters shapes must be normally distributed about a centroid.

The points of the data file are read in chunks, each chunk can be processed in main memory that consists of 3 types of objects:

- The Discard Set: Simple summaries of the clusters
- The Compressed Set: Summaries of clusters for sets of points that have been found close to one another, but not close to any cluster.
- The Retained Set: Points that can't be assigned to a cluster nor sufficiently close to any other points



K-MEANS ALGORITHMS

The BFR Data Processing:

- All points sufficiently close to the centroid of a cluster are added to that cluster.
- The points that are not sufficiently close to any centroid, we cluster them, along with the points in the retained set. The points alone is there cluster are kept in the retained set and the cluster of more than one point are added to the compressed set
- The little cluster from the compressed set are merged with the bigger cluster of the discard set if they are close enough from one of these cluster.
- The centroid of the clusters in the discard set are recomputed. If there is some points from the retained set getting close enough from each other, they are added to a small cluster in the compressions set.

K-MEANS ALGORITHMS

The Mahalanobis distance is the distance between a point and the centroid of a cluster, normalized by the standard deviation of the cluster in each dimension.

$$\sqrt{\sum_{i=1}^d \left(\frac{p_i - c_i}{\sigma_i} \right)^2}$$

p_i : the point i

c_i : the centroid i

σ_i : the standard deviation of point in the cluster in the i -th dimension

It's used to evaluate if a point is far from a cluster's centroid or not.

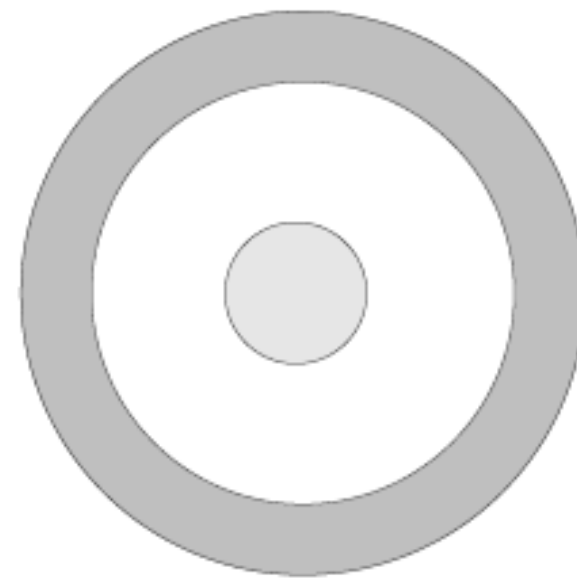


THE CURE ALGORITHM

“Clustering Using REpresentatives” (CURE) algorithm:

This algorithm assumes a Euclidean space, but it doesn't assume anything about the shape of clusters. Instead of representing clusters by their centroid, it uses a collection of representing points.

Exemple :



Here a cluster is surrounding another one

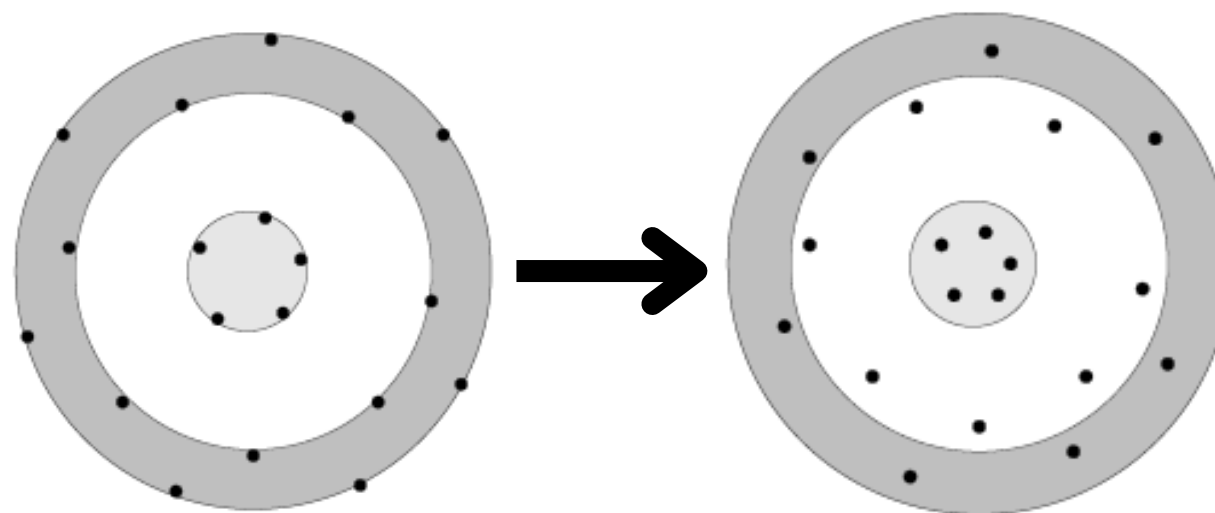
THE CURE ALGORITHM

Initialization in CURE:

- Take a small sample of the data and cluster it in main memory.
- Select a small set of points from each cluster to be “representative points”, These points should be chosen to be as far as possible from one another.
- Move each of the representative points a fixed fraction of the distance between its location and the centroid of its cluster (20% is a good fraction to choose).

Example :

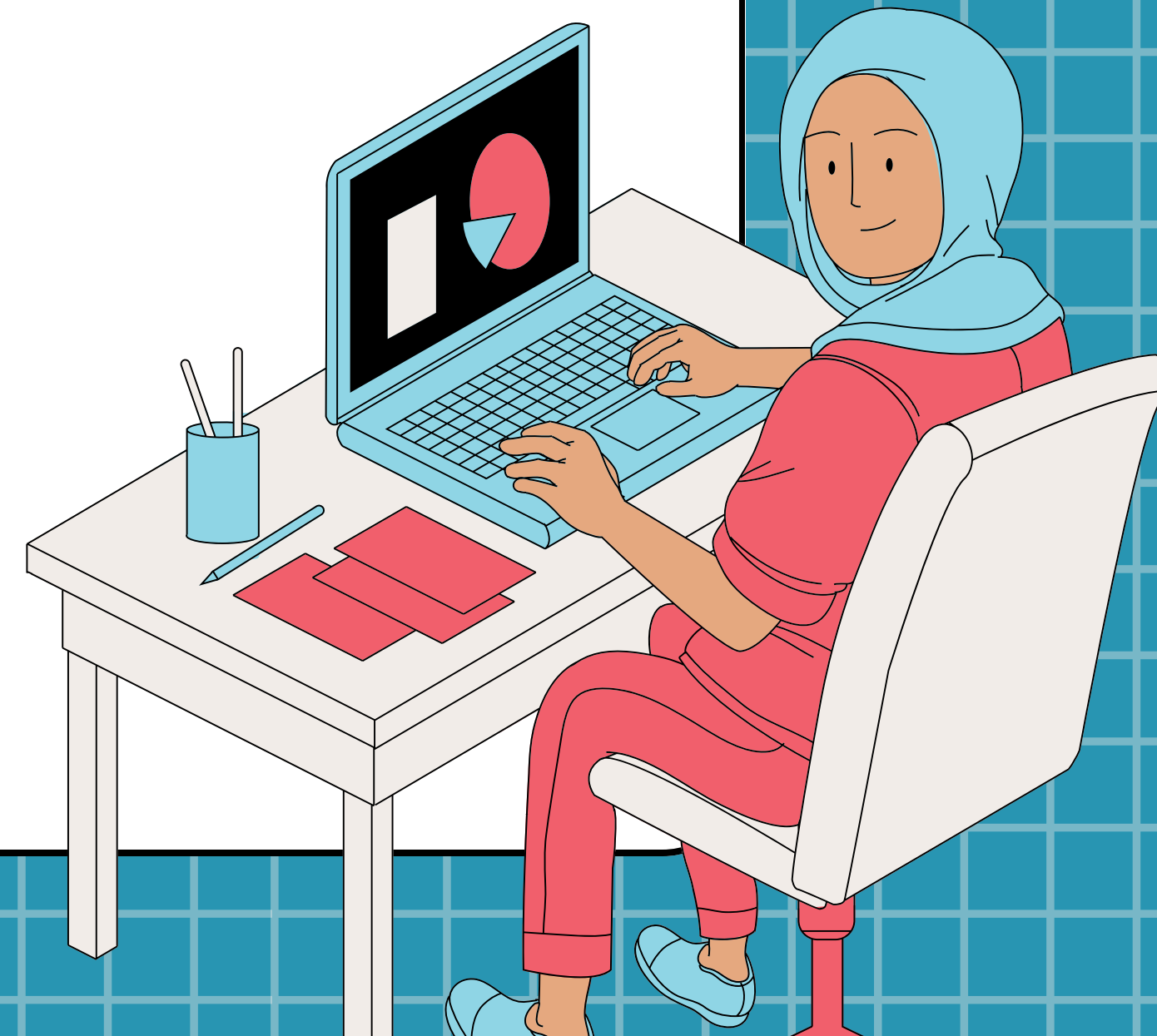
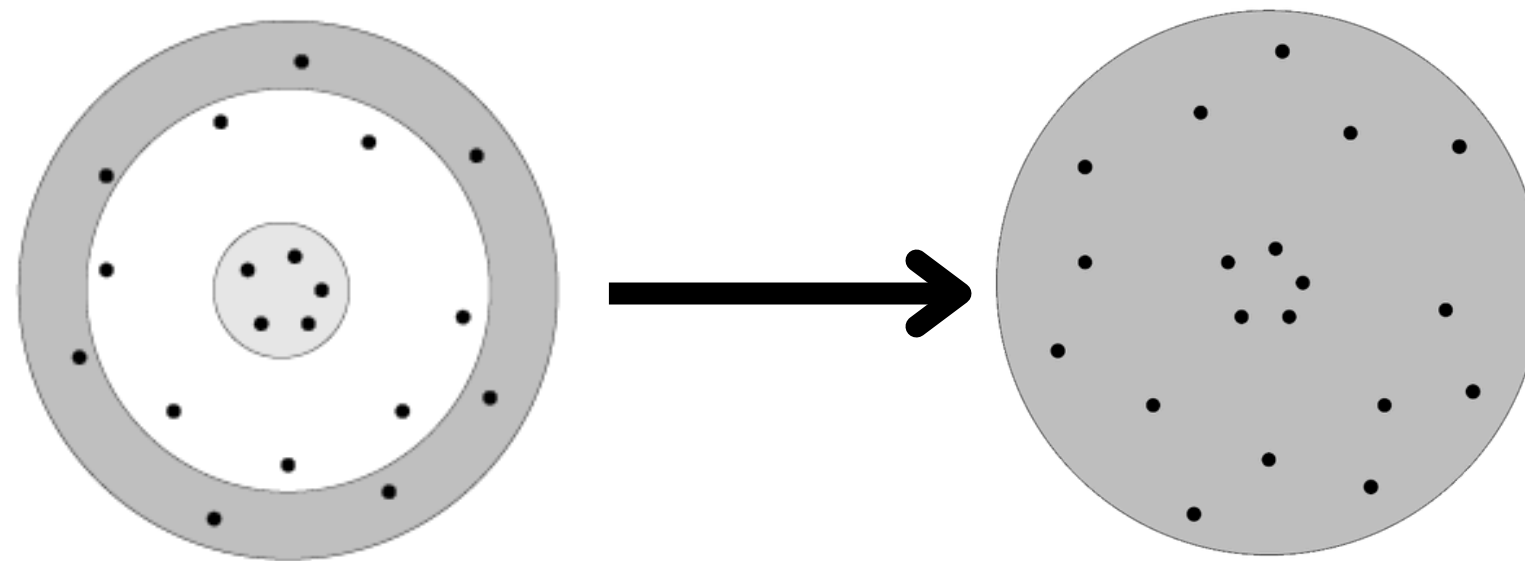
We selected representative points from each cluster, as far from one another as possible



Moving the representative points 20% of the distance to the cluster's centroid

THE CURE ALGORITHM

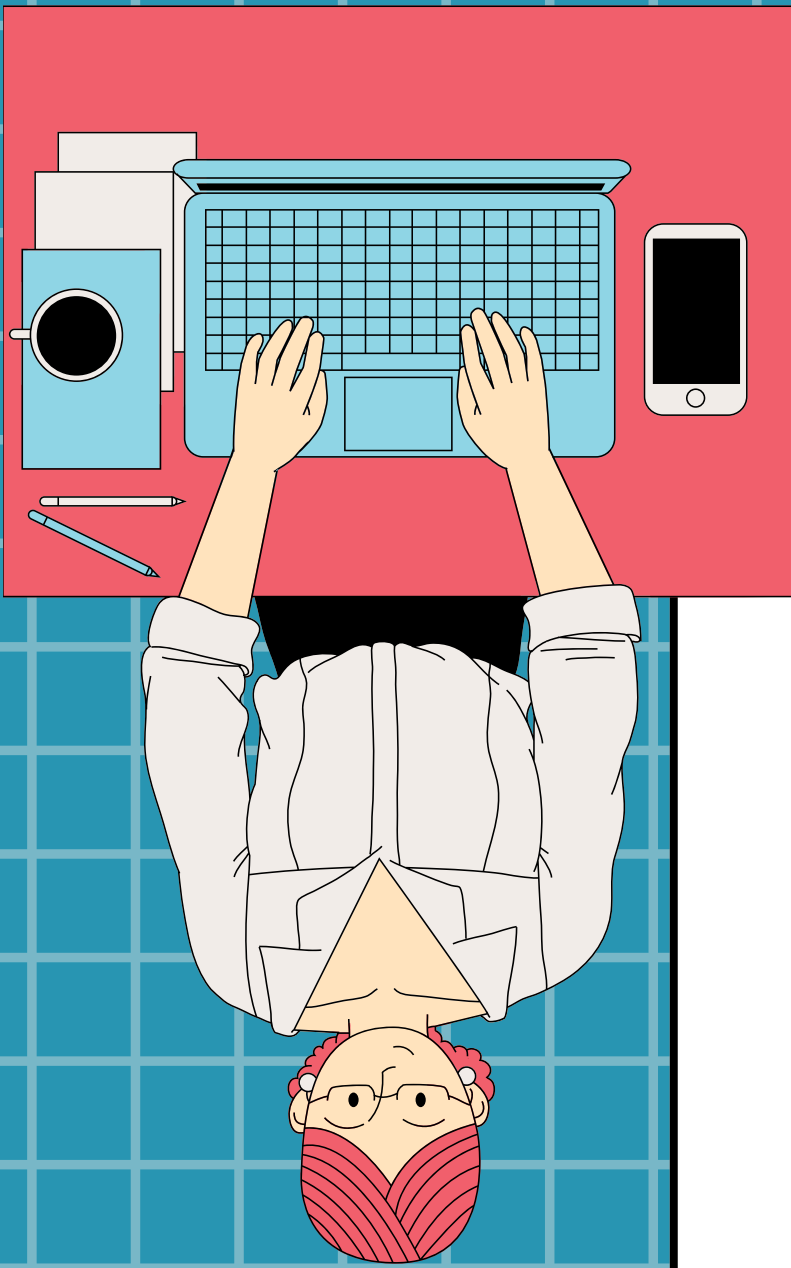
The next step is to merge clusters if they have a pair of representative points, one from each cluster. The user can define which distance can be considered “close”.



CONCLUSION

K-Means Algorithms : Family of algorithms of the point-assignment type and assuming a Euclidean space and k clusters for a known k value. The points are assigned to the closest centroid one at a time and the centroids can migrate during the point assignment.

The CURE Algorithm : This algorithm is of the point-assignment type and designed for a Euclidean space, but clusters can have any shape. It clusters few "representative points", then move them to fraction closer to the centroid and then recompute the clusters if there is any change.



**ARE THERE ANY
QUESTIONS?**

Thank for your time

